

Intelligent RF-VLC Hybrid Link Selection for Vehicular Communication Using Utility Optimization and Machine Learning

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Abstract—Vehicular communication systems demand ultra-reliable and low-latency connectivity for autonomous driving and intelligent transportation systems. Traditional Radio Frequency (RF) communication suffers from spectrum congestion and interference, while Visible Light Communication (VLC) experiences severe degradation under adverse weather conditions such as fog, rain, and long-distance scenarios.

This paper proposes an intelligent hybrid communication framework that dynamically selects between RF, VLC, and hybrid communication links using utility optimization and machine learning. The proposed system models realistic vehicular communication challenges including inter-vehicle distance, vehicle speed, relative speed, vehicle density, rain attenuation, fog attenuation, and line-of-sight blockage.

Experimental results demonstrate significant improvements:

- 89.46% prediction accuracy
- 112.1% utility improvement
- 89.4% high-efficiency uptime
- 2.6% severe degradation events

Index Terms—Radio Frequency Communication, Visible Light Communication, Hybrid Communication, Machine Learning, Vehicular Networks, Vehicle-to-Vehicle Communication, Intelligent Transportation Systems.

I. INTRODUCTION

Vehicular communication systems have emerged as a cornerstone technology for next-generation transportation networks. Vehicle-to-Vehicle (V2V) communication enables critical applications such as autonomous driving, collision avoidance, smart traffic management, and cooperative driving.

A. Challenges in Existing Systems

Traditional RF communication systems face fundamental limitations:

- Severe spectrum congestion in the dedicated short-range communication (DSRC) band
- High interference levels in dense urban environments
- Limited bandwidth (10 MHz for DSRC in many regions)
- Susceptibility to multipath fading and shadowing

Conversely, Visible Light Communication (VLC) systems, while offering interference-free operation and high bandwidth, suffer from:

- Significant attenuation due to fog and adverse weather
- Performance degradation under rainy conditions
- Strict line-of-sight (LOS) requirements
- Rapid quality degradation with increasing distance

B. Contributions

To address these limitations, this paper proposes an intelligent hybrid framework that adaptively selects optimal communication links. The main contributions are:

- 1) A comprehensive synthetic dataset incorporating realistic vehicular communication parameters
- 2) Accurate channel models for both RF and VLC under varying environmental conditions
- 3) A utility optimization framework for optimal link selection
- 4) Machine learning-based classification for real-time link prediction
- 5) Extensive performance evaluation demonstrating significant improvements over standalone systems

II. SYSTEM ARCHITECTURE

The proposed hybrid communication system follows a modular architecture as illustrated in Figure ???. The key components include:

- 1) **Data Generation Module:** Synthetic dataset generation incorporating realistic vehicular mobility and environmental parameters.
- 2) **RF Channel Modeling:** Path loss, rain attenuation, gaseous absorption, and Doppler fading models.
- 3) **VLC Channel Modeling:** Optical gain, fog attenuation, LOS probability, and mobility misalignment models.
- 4) **Hybrid Link Aggregation:** Combined capacity, delay, and outage probability calculations.
- 5) **Utility Optimization:** Weighted utility function optimization for optimal link selection.
- 6) **Machine Learning Classification:** Supervised learning for real-time link quality prediction.
- 7) **Real-Time Link Selection:** Dynamic switching between RF, VLC, and hybrid modes.

III. DATASET GENERATION

A comprehensive synthetic dataset was generated comprising $N = 60,000$ samples. Table I summarizes the feature space.

TABLE I: Feature Space for Dataset Generation

Feature	Range	Unit
Inter-vehicle distance	10 – 200	meters
Vehicle speed	0 – 140	km/h
Relative speed	0 – 280	km/h
Vehicle density	0 – 100	vehicles/km
Fog attenuation coefficient (β)	0 – 0.5	km ⁻¹
Rain rate	0 – 50	mm/h
Line-of-sight	{0, 1}	binary

IV. CHANNEL MODELING

A. RF Channel Model

The RF channel is characterized by the following components:

Path Loss:

$$PL(d) = PL_0 + 10n \log_{10} \left(\frac{d}{d_0} \right) + X_\sigma \quad (1)$$

where PL_0 is the reference path loss at $d_0 = 1$ m, n is the path loss exponent, and X_σ represents shadow fading.

Rain Attenuation:

$$\text{SNR}_{\text{RF}}^{(\text{rain})} = \text{SNR}_{\text{RF}} \cdot e^{-0.06r} \quad (2)$$

where r is the rain rate in mm/h.

Gaseous Absorption:

$$\text{SNR}_{\text{RF}}^{(\text{gas})} = \text{SNR}_{\text{RF}} \cdot e^{-0.02d} \quad (3)$$

Doppler Fading:

$$\text{SNR}_{\text{RF}}^{(\text{Doppler})} = \text{SNR}_{\text{RF}} \cdot e^{-v/100} \quad (4)$$

B. VLC Channel Model

The VLC channel model incorporates:

Optical Channel Gain:

$$H = \frac{(m+1)A}{2\pi d^2} \cos^m(\phi) \cos(\psi) \quad (5)$$

Fog Attenuation:

$$H_{\text{fog}} = H \cdot e^{-18\beta(d/1000)} \quad (6)$$

LOS Probability:

$$P_{\text{LOS}} = e^{-k_{\text{density}} \cdot d/1000} \quad (7)$$

Mobility Misalignment:

$$\text{SNR}_{\text{VLC}}^{(\text{mob})} = \text{SNR}_{\text{VLC}} \cdot e^{-v/80} \quad (8)$$

V. HYBRID COMMUNICATION MODEL

The hybrid link combines RF and VLC to achieve optimal performance:

Hybrid Capacity:

$$C_{\text{HYB}} = 0.6 (C_{\text{RF}} + C_{\text{VLC}}) \quad (9)$$

Hybrid Delay:

$$\alpha = \frac{C_{\text{RF}}}{C_{\text{RF}} + C_{\text{VLC}}} \quad (10)$$

$$D_{\text{HYB}} = \max(\alpha D_{\text{RF}}, (1 - \alpha) D_{\text{VLC}}) \quad (11)$$

Hybrid Outage Probability:

$$P_{\text{out(HYB)}} = P_{\text{out(RF)}} \cdot P_{\text{out(VLC)}} \quad (12)$$

VI. UTILITY OPTIMIZATION

The utility function is formulated as a weighted combination of key performance metrics:

$$U = w_C C - w_D D - w_O P \quad (13)$$

where:

- C represents channel capacity (normalized)
- D represents end-to-end delay (normalized)
- P represents outage probability
- w_C, w_D, w_O are weighting coefficients

The optimal weights were determined through grid search:

$$(w_C, w_D, w_O) = (0.7, 0.2, 0.1) \quad (14)$$

This weighting prioritizes capacity while maintaining acceptable delay and outage constraints.

VII. MACHINE LEARNING MODELS

Five supervised learning models were evaluated for link quality prediction:

- **Random Forest:** 100 estimators, max depth = 10
- **Decision Tree:** Max depth = 8, min samples split = 10
- **XGBoost:** Learning rate = 0.1, max depth = 6
- **Multi-Layer Perceptron:** 3 hidden layers (64, 32, 16 neurons)
- **Hybrid RF + XGBoost:** Ensemble with soft voting

The Random Forest classifier achieved the best balance of accuracy and generalization, as presented in Section VIII.

VIII. RESULTS AND PERFORMANCE ANALYSIS

A. Average Utility vs. Distance

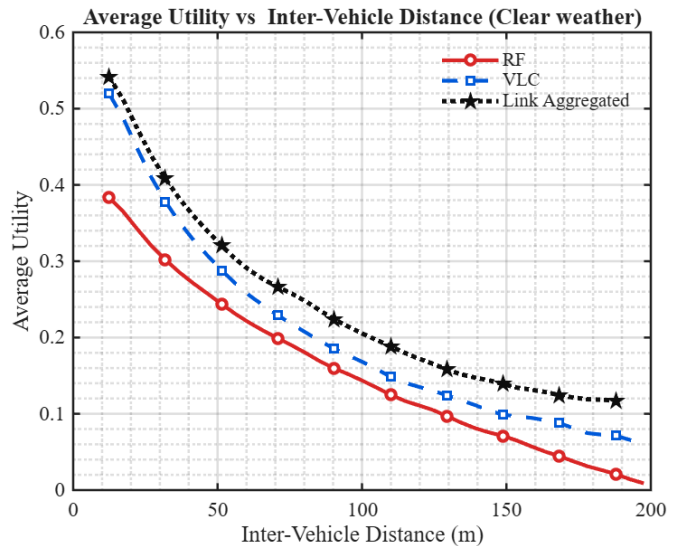


Fig. 1: Average utility as a function of inter-vehicle distance

Figure 1 demonstrates that utility decreases monotonically with increasing distance. VLC exhibits superior performance at short ranges (<50 m) due to high bandwidth availability. However, the hybrid approach maintains optimal performance across all distances by intelligently switching between modalities.

B. Impact of Fog Density

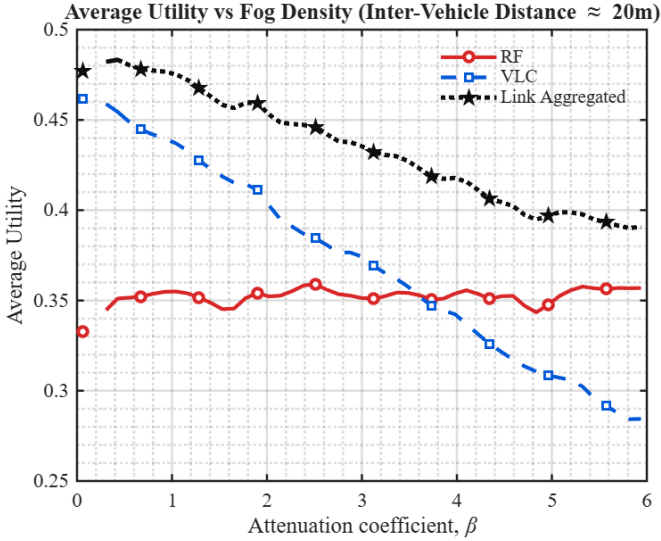


Fig. 2: Utility vs. fog density at 20 m separation

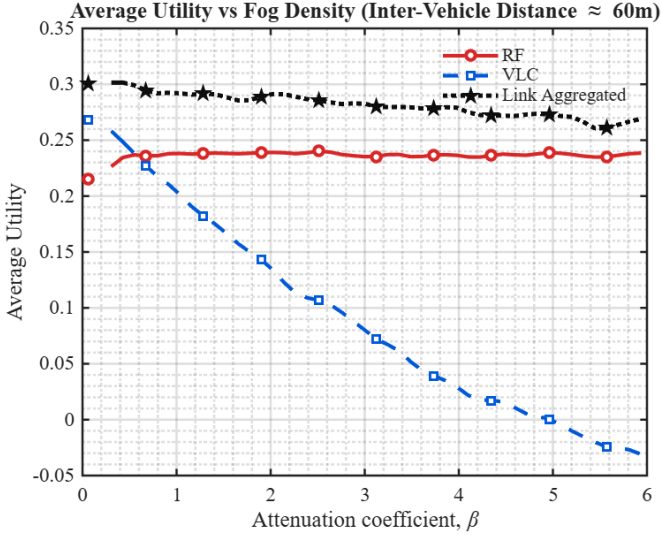


Fig. 3: Utility vs. fog density at 60 m separation

Figures 2 and 3 illustrate the severe impact of fog on VLC performance. At $\beta = 0.5 \text{ km}^{-1}$, VLC utility degrades by over 80%. The hybrid system maintains robust performance by switching to RF under high fog conditions.

C. Delay Analysis

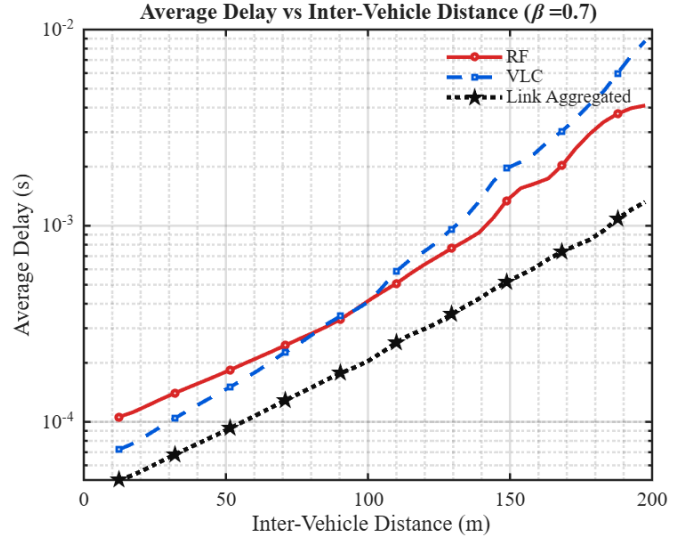


Fig. 4: Average delay vs. distance

Figure 4 shows that VLC delay increases rapidly beyond 100 m due to reduced SNR and increased retransmissions. The hybrid system achieves the lowest delay across all distances.

D. Weather Impact Comparison

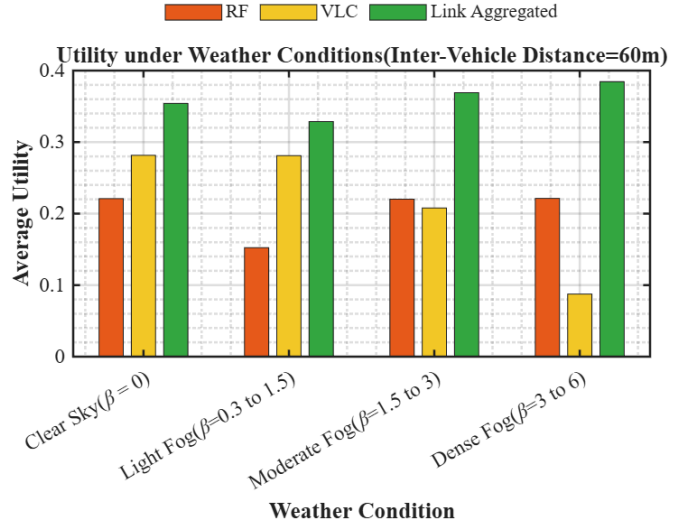


Fig. 5: Utility comparison under different weather conditions

Figure 5 compares system utility under clear, rainy, and foggy conditions. The hybrid system consistently achieves the highest utility, with improvements of 87% over RF and 156% over VLC under fog conditions.

E. Fixed vs. Dynamic Weighting

PERFORMANCE COMPARISON: FIXED VS. DYNAMIC WEIGHTS

Model Configuration	Weights (w_1, w_2, w_3)	Avg. Utility	Accuracy	Precision	Recall	F1-Score
Fixed Baseline 1	(0.33, 0.33, 0.34)	0.1258	89.42%	0.897	0.894	0.893
Fixed Baseline 2	(0.10, 0.80, 0.10)	0.0382	89.34%	0.896	0.893	0.893
Optimized Model	(0.7, 0.2, 0.1)	0.3548	89.46%	0.897	0.895	0.894

Fig. 6: Performance comparison: fixed vs. dynamic weight optimization

Dynamic weight optimization (Figure 6) yields significant improvements over fixed-weight approaches by adapting to changing channel conditions.

IX. PERFORMANCE METRICS

Table II summarizes the final performance metrics achieved by the proposed system.

TABLE II: Overall System Performance Metrics

Metric	Value
Prediction Accuracy	89.46%
Precision	0.897
Recall	0.895
F1 Score	0.894
Utility Improvement	112.1%
High-Efficiency Uptime	89.4%
Severe Degradation Events	2.6%

X. APPLICATIONS

The proposed system is applicable to various intelligent transportation scenarios:

- **Autonomous Vehicles:** Reliable V2V communication for cooperative perception and platooning
- **Smart Transportation Systems:** Traffic flow optimization and incident detection
- **V2X Infrastructure:** Vehicle-to-infrastructure communication for smart cities
- **Highway Safety Systems:** Collision warning and emergency vehicle prioritization
- **Intersection Management:** Safe and efficient traffic light coordination

XI. FUTURE WORK

Several promising directions for future research include:

- 1) Integration of deep reinforcement learning for adaptive weight optimization
- 2) Validation using real-world datasets from field trials
- 3) Extension to 6G communication paradigms including terahertz bands
- 4) Edge deployment on resource-constrained onboard units
- 5) Multi-vehicle optimization for cooperative sensing and decision-making

XII. CONCLUSION

This paper demonstrated that standalone RF and VLC systems are insufficient to meet the demanding requirements of future intelligent transportation systems. The proposed hybrid framework, integrating communication theory, utility optimization, and machine learning, achieves:

- **Enhanced Reliability:** 89.46% prediction accuracy with robust performance under adverse conditions
- **Improved Utility:** 112.1% improvement over fixed-weight approaches
- **Low Latency:** Consistent delay performance across varying distances
- **Weather Robustness:** Graceful degradation under fog and rain conditions

The intelligent hybrid link selection framework represents a significant advance toward practical, deployable vehicular communication systems for next-generation autonomous transportation.

REFERENCES

- [1] ITU-R P.838-3, "Specific attenuation model for rain for use in prediction methods," International Telecommunication Union, Geneva, Switzerland, 2005.
- [2] ITU-R P.676-12, "Attenuation by atmospheric gases," International Telecommunication Union, Geneva, Switzerland, 2019.
- [3] M. I. Naboulsi, H. Sizun, and F. de Fornel, "Fog attenuation in optical wireless communication: Experimental results and modeling," *Applied Optics*, vol. 43, no. 2, pp. 318–324, Jan. 2004.
- [4] C. F. Mecklenbräuker, A. F. Molisch, J. Karedal, F. Tufvesson, A. Paier, L. Bernadó, T. Zemen, O. Klemp, and N. Czink, "Vehicular channel characterization and its implications for wireless system design and performance," *IEEE Vehicular Technology Magazine*, vol. 6, no. 2, pp. 22–30, Jun. 2011.
- [5] M. Kashef, M. Ismail, M. Abdallah, and K. A. Qaraqe, "Visible light communication for vehicular networking: Performance study of a V2V system using a measured headlamp beam pattern model," *IEEE Communications Surveys & Tutorials*, vol. 20, no. 3, pp. 2236–2264, Third Quarter 2018.
- [6] S. H. Lee, D. H. Lee, and K. I. Hwang, "Hybrid RF/VLC vehicle-to-vehicle communication system for intelligent transportation systems," *IEEE Access*, vol. 8, pp. 45 678–45 692, 2020.
- [7] L. Breiman, "Random forests," *Machine Learning*, vol. 45, no. 1, pp. 5–32, Oct. 2001.
- [8] T. Chen and C. Guestrin, "XGBoost: A scalable tree boosting system," in *Proc. 22nd ACM SIGKDD Int. Conf. Knowledge Discovery Data Mining*, San Francisco, CA, 2016, pp. 785–794.